



IPv6 Basics



Foreword

- In the 1980s, the Internet Engineering Task Force (IETF) released RFC 791 – Internet Protocol, which marks the standardization of IPv4. In the following decades, IPv4 has become one of the most popular protocols. Numerous people have developed various applications based on IPv4 and made various supplements and enhancements to IPv4, enabling the Internet to flourish.
- However, with the expansion of the Internet and the development of new technologies such as 5G and Internet of Things (IoT), IPv4 faces more and more challenges. It is imperative to replace IPv4 with IPv6.
- This course describes the reasons for IPv4-to-IPv6 transition and basic IPv6 knowledge.



Objectives

- On completion of this course, you will be able to:
 - Summarize the advantages of IPv6 over IPv4.
 - Describe the basic concepts of IPv6.
 - Describe the formats and functions of IPv6 packet headers.
 - Describe the IPv6 address format and address types.
 - Describe the method and basic procedure for configuring IPv6 addresses.
 - Configure IPv6 addresses and IPv6 static routes.



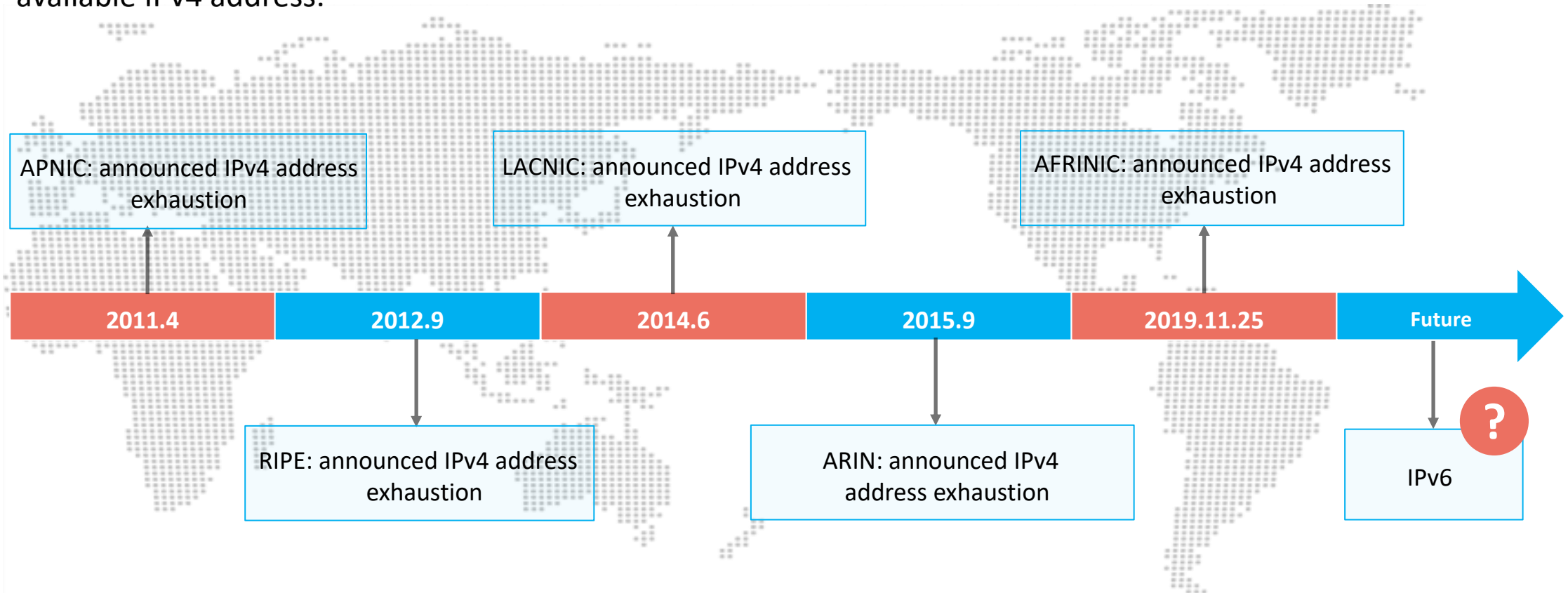
Contents

- 1. IPv6 Overview**
2. IPv6 Address Configuration
3. Typical IPv6 Configuration Examples



IPv4 Status

- On February 3, 2011, the Internet Assigned Numbers Authority (IANA) announced even allocation of its last 4.68 million IPv4 addresses to five Regional Internet Registries (RIRs) around the world. The IANA thereafter had no available IPv4 address.





Why IPv6?

IPv4

Exhausted public IP addresses

Improper packet header design

Large routing table, leading to inefficient
table query

Dependency on ARP causes broadcast
storms

...

vs.

IPv6

Nearly infinite address space

Hierarchical address allocation

Plug-and-play

Simplified packet header

IPv6 security features

Integrity of E2E communication

Support for mobility

Enhanced QoS features

...



IPv6 Advantages

Nearly infinite address space

The 128-bit address length provides numerous addresses, meeting the requirements of emerging services such as the IoT and facilitating service evolution and expansion.

Hierarchical address structure

IPv6 addresses are allocated more properly than IPv4 addresses, facilitating route aggregation (reducing the size of IPv6 routing tables) and fast route query.

Plug-and-play

IPv6 supports stateless address autoconfiguration (SLAAC), simplifying terminal access.

Simplified packet header

The simplified packet header improves forwarding efficiency. New applications can be supported using extension headers, which facilitate the forwarding processing of network devices and reduce investment costs.

Security features

IPsec, source address authentication, and other security features ensure E2E security, preventing NAT from damaging the integrity of E2E communication.

Mobility

Greatly improves real-time communication and performance of mobile networks.

Enhanced QoS features

A Flow Label field is additionally defined and can be used to allocate a specific resource for a special service and data flow.



Basic IPv6 Header

- An IPv6 header consists of a mandatory basic IPv6 header and optional extension headers.
- The basic header provides basic information for packet forwarding and is parsed by all devices on a forwarding path.

IPv4 packet header (20–60 bytes)

Version	HL	ToS	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Head Checksum	
Source Address				
Destination Address				
Options				Padding

Basic IPv6 header (40 bytes)

Version	Traffic Class	Flow Label		
Payload Length		Next Header	Hop Limit	
Source Address				
Destination Address				

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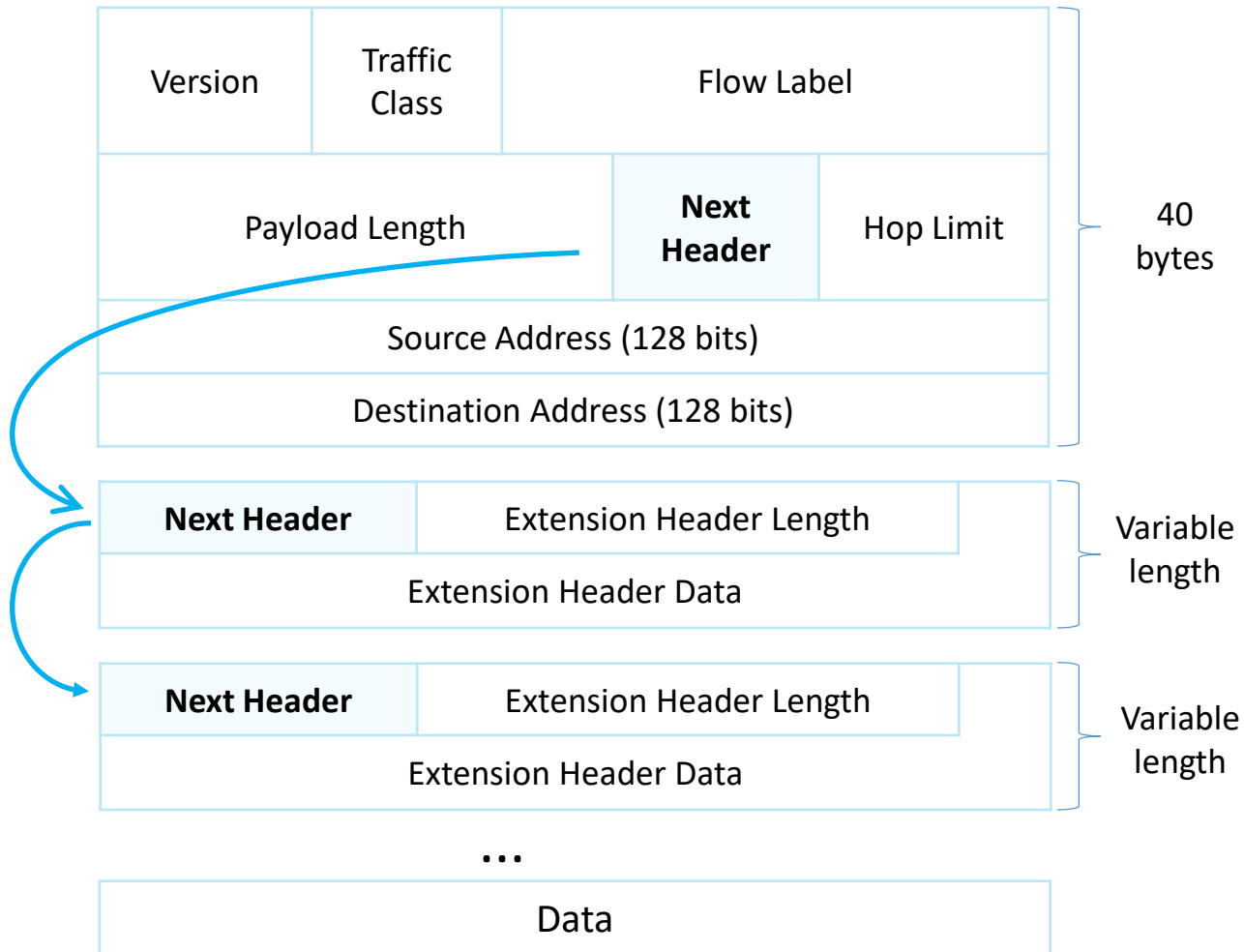
Reserved

Name/Location
changed

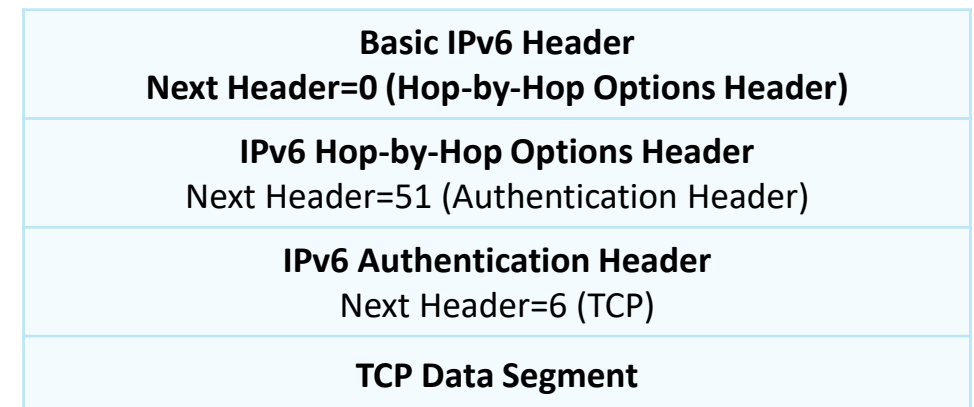
New



IPv6 Extension Header



- **Extension Header Length:** 8 bits long. This field indicates the extension header length excluding the length of the Next Header field.
- **Extension Header Data:** variable length. This field indicates the payload of the extension headers and is a combination of a series of options and padding fields.



IPv6 packet example



IPv6 Packet Processing Mechanism

Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header=51 (Authentication Header)
IPv6 Authentication Header Next Header=6 (TCP)
TCP Data Segment

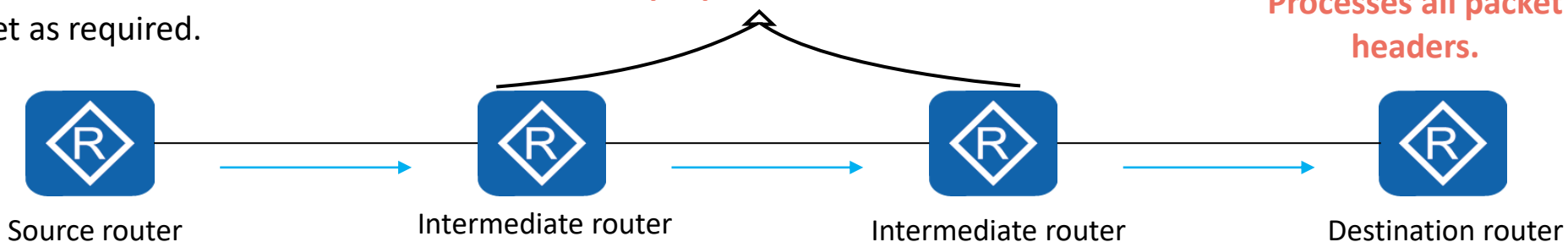
Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
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TCP Data Segment

Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header=51 (Authentication Header)
IPv6 Authentication Header Next Header=6 (TCP)
TCP Data Segment

Constructs an IPv6 packet as required.

Process the **basic header** and **Hop-by-Hop Options header**.

Processes all packet headers.

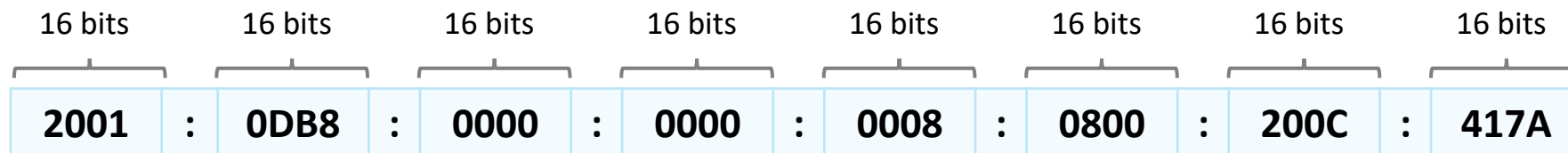


- The length of the basic packet header is fixed, **improving the forwarding efficiency**.
- The extension headers **meet special requirements**.



IPv6 Address

- The length of an IPv6 address is 128 bits. Colons are generally used to divide the IPv6 address into eight segments. Each segment contains 16 bits and is expressed in hexadecimal notation.



The letters in an IPv6 address are case insensitive. For example, **A** is equivalent to **a**.

- Similar to an IPv4 address, an IPv6 address is expressed in the format of IPv6 address/mask length.
 - Example: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF/64

IPv6 address: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF

Subnet number: 2001:0DB8:2345:CD30::/64



IPv6 Address Abbreviation Specifications

- For convenience, IPv6 can be abbreviated according to the following rules.

Abbreviation Specifications

2001	:	0DB8	:	0000	:	0000	:	0008	:	0800	:	200C	:	417A
------	---	------	---	------	---	------	---	------	---	------	---	------	---	------

The leading 0s in each 16-bit segment can be omitted. However, if all bits in a 16-bit segment are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted.



2001	:	DB8	:	0	:	0	:	8	:	800	:	200C	:	417A
------	---	-----	---	---	---	---	---	---	---	-----	---	------	---	------

If one or more consecutive 16-bit segments contain only 0s, a double colon (::) can be used to represent them, but only one :: is allowed in an entire IPv6 address.



2001	:	DB8	:	::	:	8	:	800	:	200C	:	417A
------	---	-----	---	----	---	---	---	-----	---	------	---	------

If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.

Abbreviation Examples

Before 0000:0000:0000:0000:0000:0000:0000:0001 After
::1

Before 2001:0DB8:0000:0000:FB00:1400:5000:45FF After
2001:DB8::FB00:1400:5000:45FF

Before 2001:0DB8:0000:0000:0000:2A2A:0000:0001 After
2001:DB8::2A2A:0:1

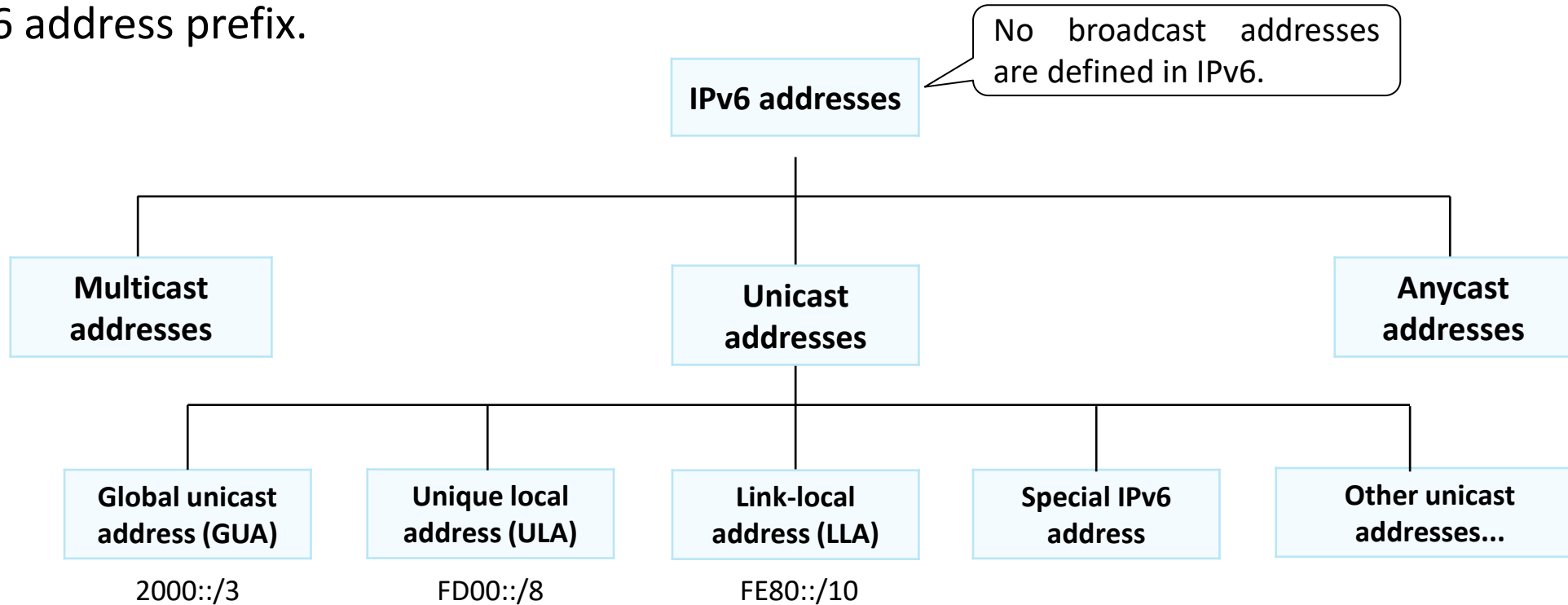
Before 2001:0DB8:0000:1234:FB00:0000:5000:45FF After
2001:DB8::1234:FB00:0:5000:45FF

or 2001:DB8:0:1234:FB00::5000:45FF



IPv6 Address Classification

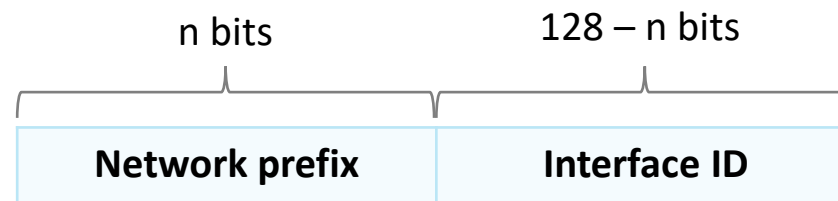
- IPv6 addresses are classified into unicast, multicast, and anycast addresses according to the IPv6 address prefix.





IPv6 Unicast Address Format

- An IPv6 unicast address is composed of two parts:
 - Network prefix: consists of n bits and is parallel to the network ID of an IPv4 address.
 - Interface ID: consists of $(128 - n)$ bits and is parallel to the host ID of an IPv4 address.
- Common IPv6 unicast addresses, such as GUAs and LLAs, require that the network prefix and interface ID be 64 bits.





Interface ID of an IPv6 Unicast Address

- 3 methods to generate an interface ID:
 - Manual configuration
 - Automatic generation by the system
 - Using the IEEE 64-bit extended unique identifier (EUI-64) standard
- EUI-64 is most commonly used. It converts the MAC address of an interface into an IPv6 interface ID.

MAC address (hexadecimal) 3C-52-82-49-7E-9D

MAC address (binary) 00111100-10010010-10000010 - 01001001-01111110-10011101

1 Bit 7 inversion

2 Insert FFFE

EUI-64 ID
(binary)

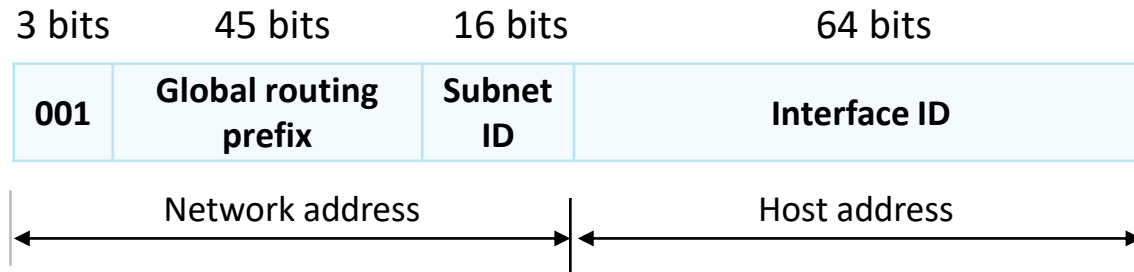
00111110-10010010-10000010-11111111-11111110-01001001-01111110-10011101

EUI-64 ID (hexadecimal) 3E-52-82-FF-FE-49-7E-9D

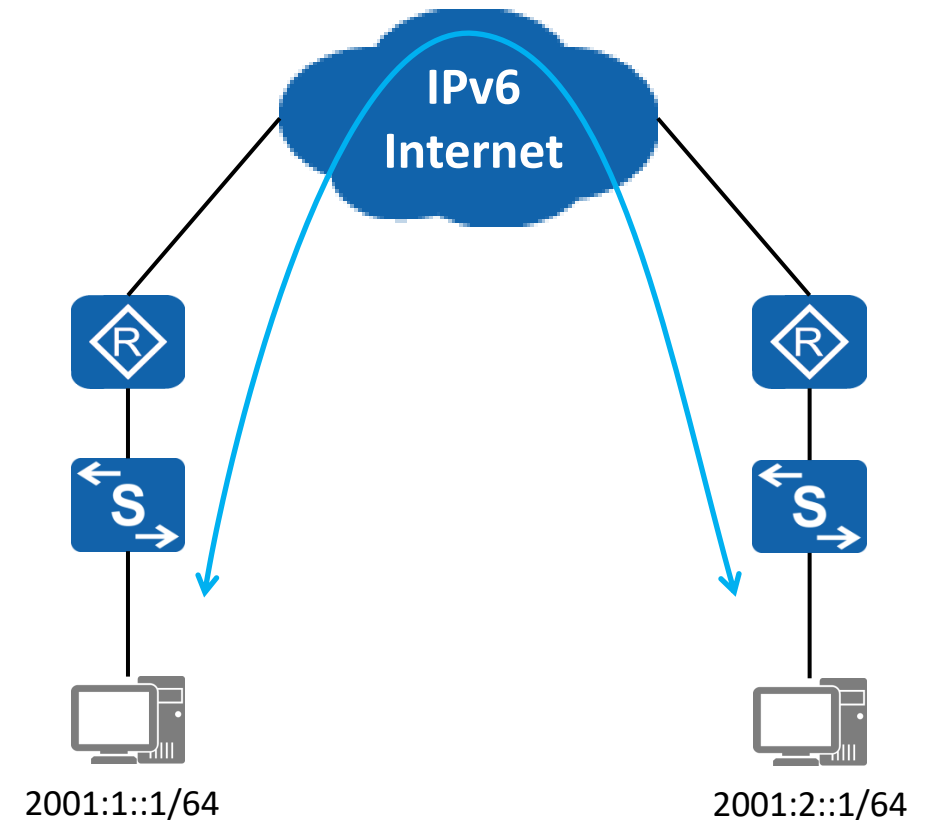


Common IPv6 Unicast Address - GUA

- A GUA is also called an aggregatable GUA. This type of address is globally unique and is used by hosts that need to access the Internet. It is equivalent to a public IPv4 address.



- The network address and interface ID of a GUA are each generally 64 bits long.
- Global routing prefix: is assigned by a provider to an organization and is generally at least 45 bits.
- Subnet ID: An organization can divide subnets based on network requirements.
- Interface ID: identifies a device's interface.





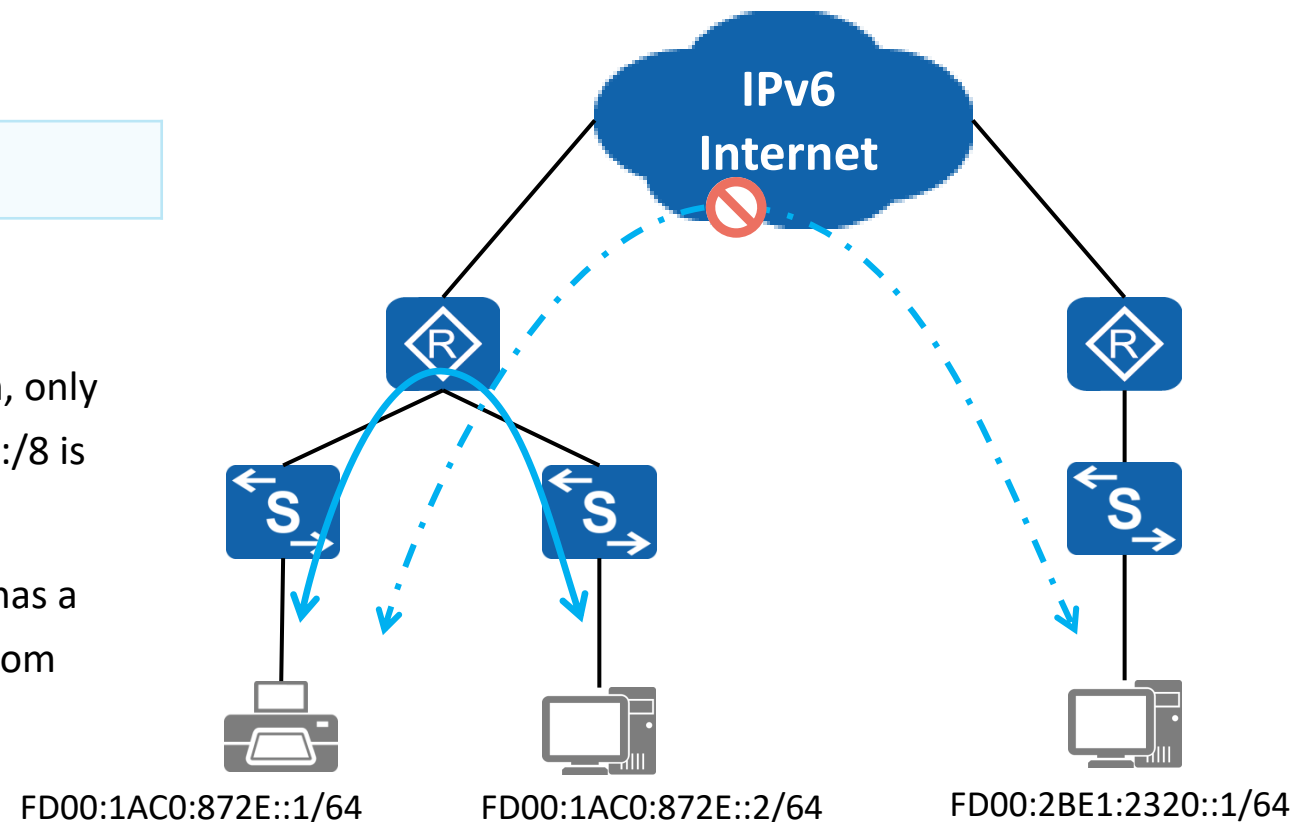
Common IPv6 Unicast Address - ULA

- A ULA is a private IPv6 address that can be used only on an intranet. This type of address cannot be routed on an IPv6 public network and therefore cannot be used to directly access a public network.

8 bits	40 bits	16 bits	64 bits
1111 1101	Global ID	Subnet ID	Interface ID

Generated using a
pseudo-random algorithm

- ULAs use the FC00::/7 address segment, among which, only the FD00::/8 address segment is currently used. FC00::/8 is reserved for future expansion.
- Although a ULA is valid only in a limited range, it also has a globally unique prefix (generated using a pseudo-random algorithm, low conflict probability).



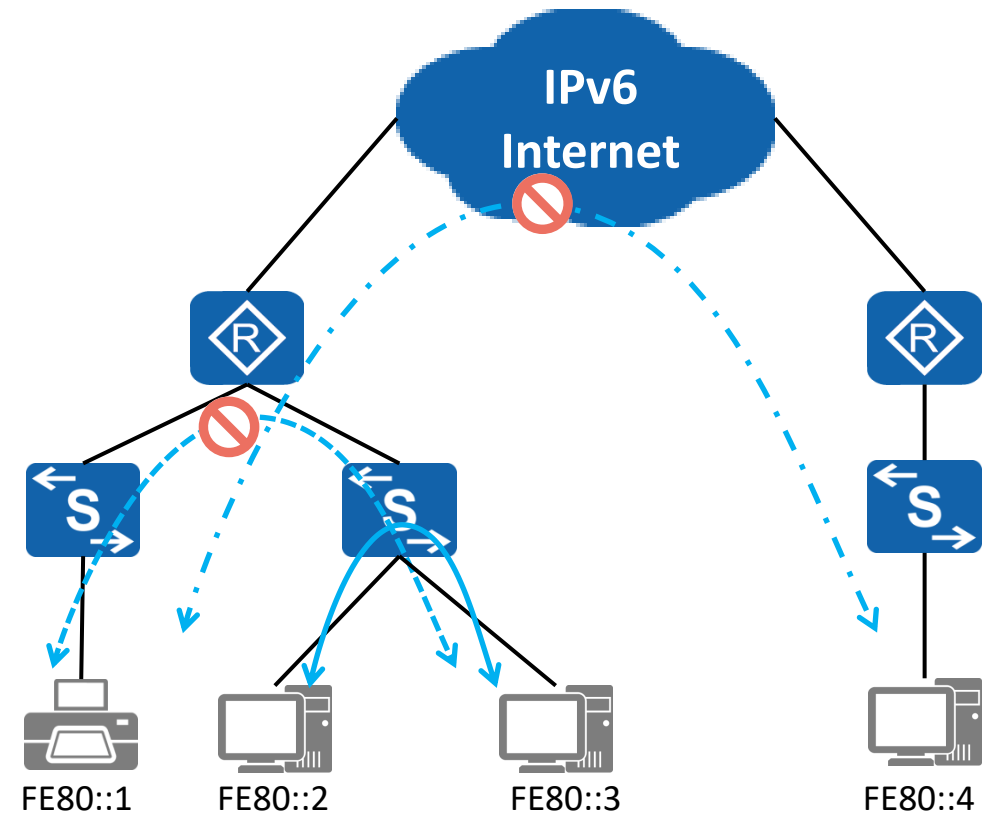


Common IPv6 Unicast Address - LLA

- An LLA is another type of IPv6 address with limited application scope. The valid range of the LLA is the local link, with the prefix of FE80::/10.

10 bit	54 bit	64 bit
1111 1110 10	0	Interface ID
Fixed at 0		

- An LLA is used for communication on a single link, such as during IPv6 SLAAC and IPv6 neighbor discovery.
- Data packets with the source or destination IPv6 address being an LLA are not forwarded out of the originating link. In other words, the valid scope of an LLA is the local link.
- Each IPv6 interface must have an LLA. Huawei devices support automatic generation and manual configuration of LLAs.



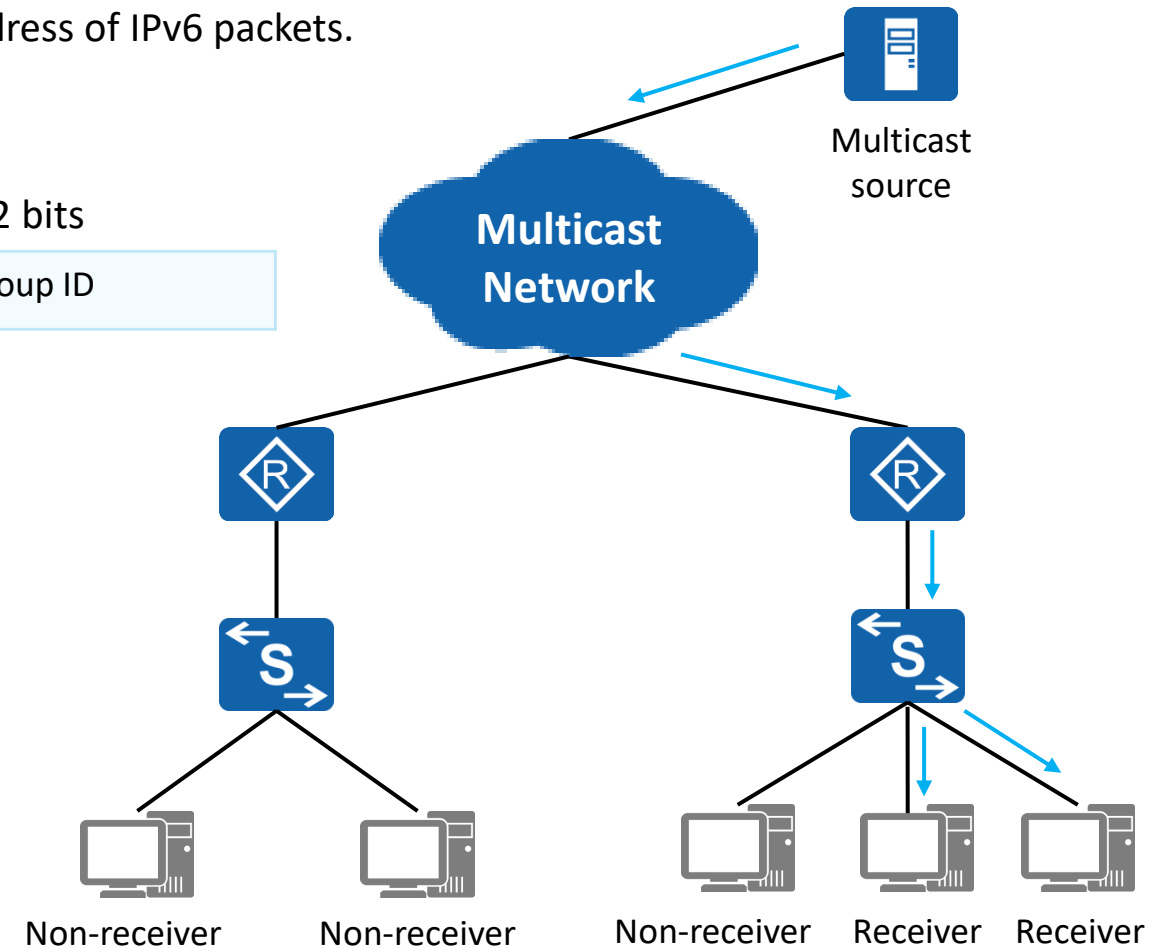


IPv6 Multicast Address

- An IPv6 multicast address identifies multiple interfaces and is generally used in one-to-many communication scenarios.
- An IPv6 multicast address can be used only as the destination address of IPv6 packets.

8 bits	4 bits	4 bits	80 bits	32 bits
11111111	Flags	Scope	Reserved (must be 0)	Group ID

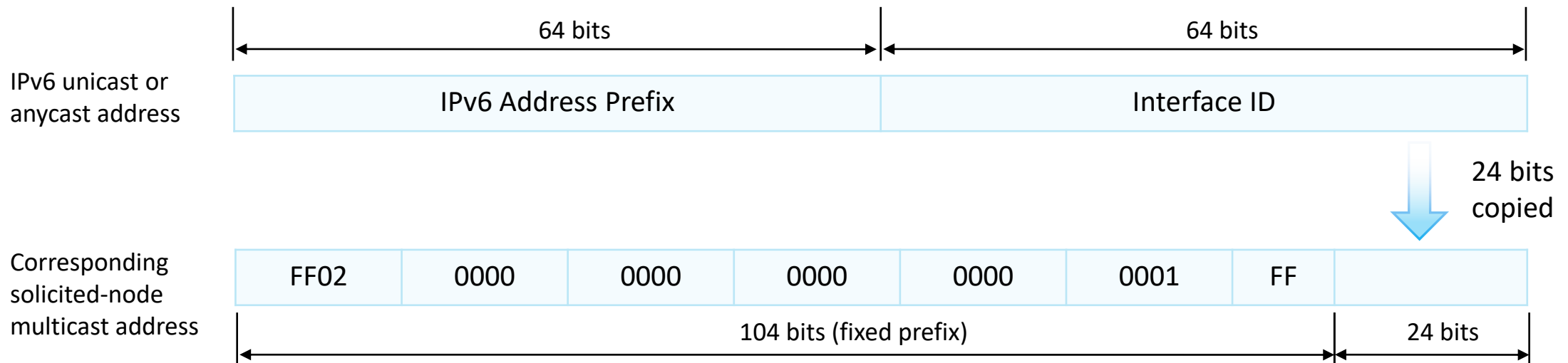
- **Flags:** indicates a permanent or transient multicast group.
- **Scope:** indicates the multicast group scope.
- **Group ID:** indicates a multicast group ID.





Solicited-Node Multicast Address

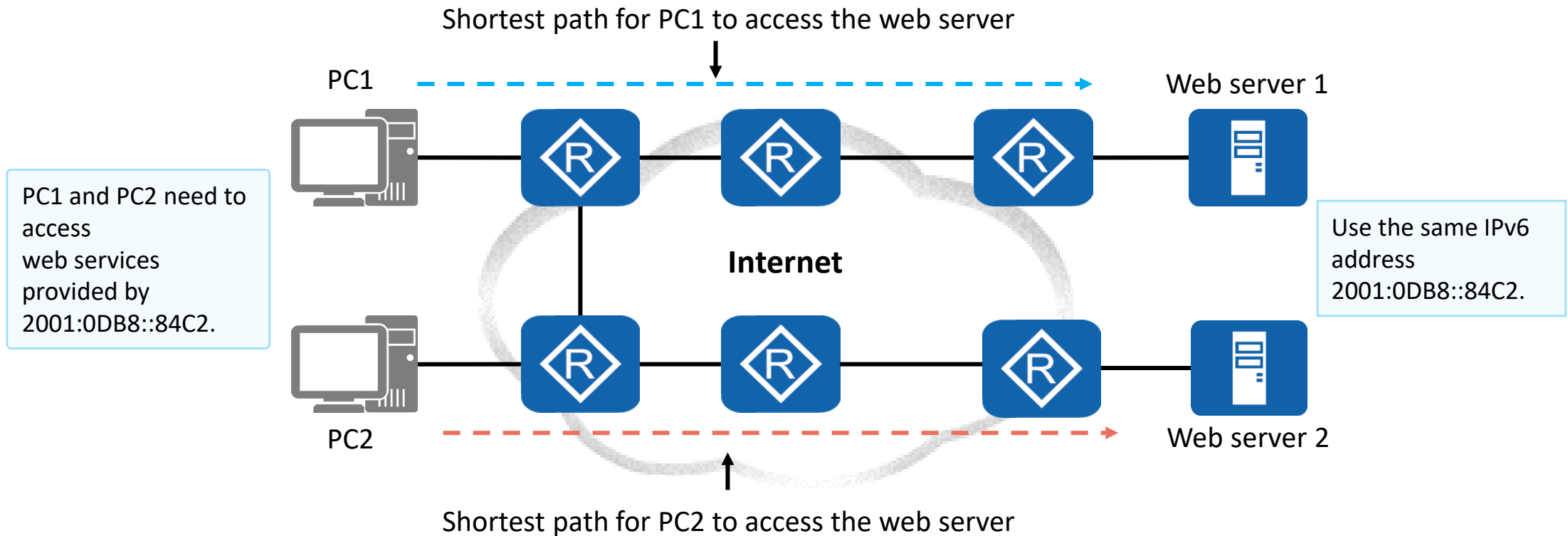
- If a node has an IPv6 unicast or anycast address, a solicited-node multicast address is generated for the address, and the node joins the corresponding multicast group. This address is used for **neighbor discovery** and **duplicate address detection (DAD)**. A solicited-node multicast address is **valid only on the local link**.





IPv6 Anycast Address

- An anycast address identifies a group of network interfaces, which usually belong to different nodes. An anycast address can be used as the source or destination address of IPv6 packets.





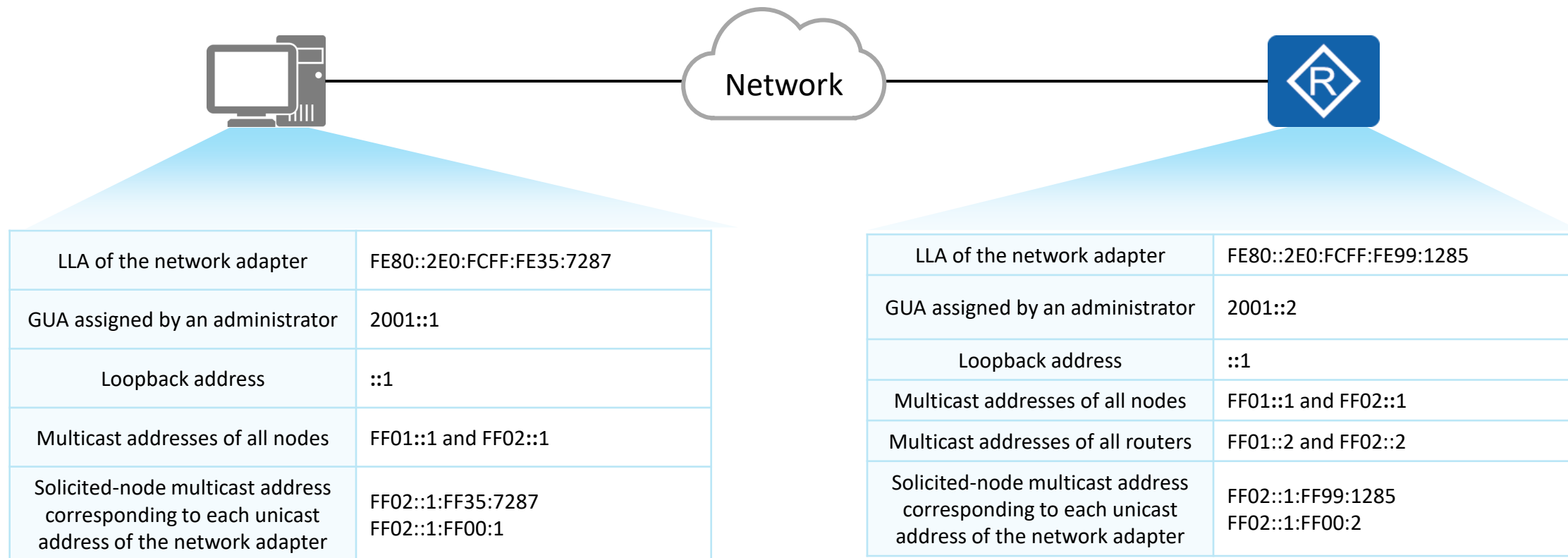
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IPv6 Addresses of Hosts and Routers

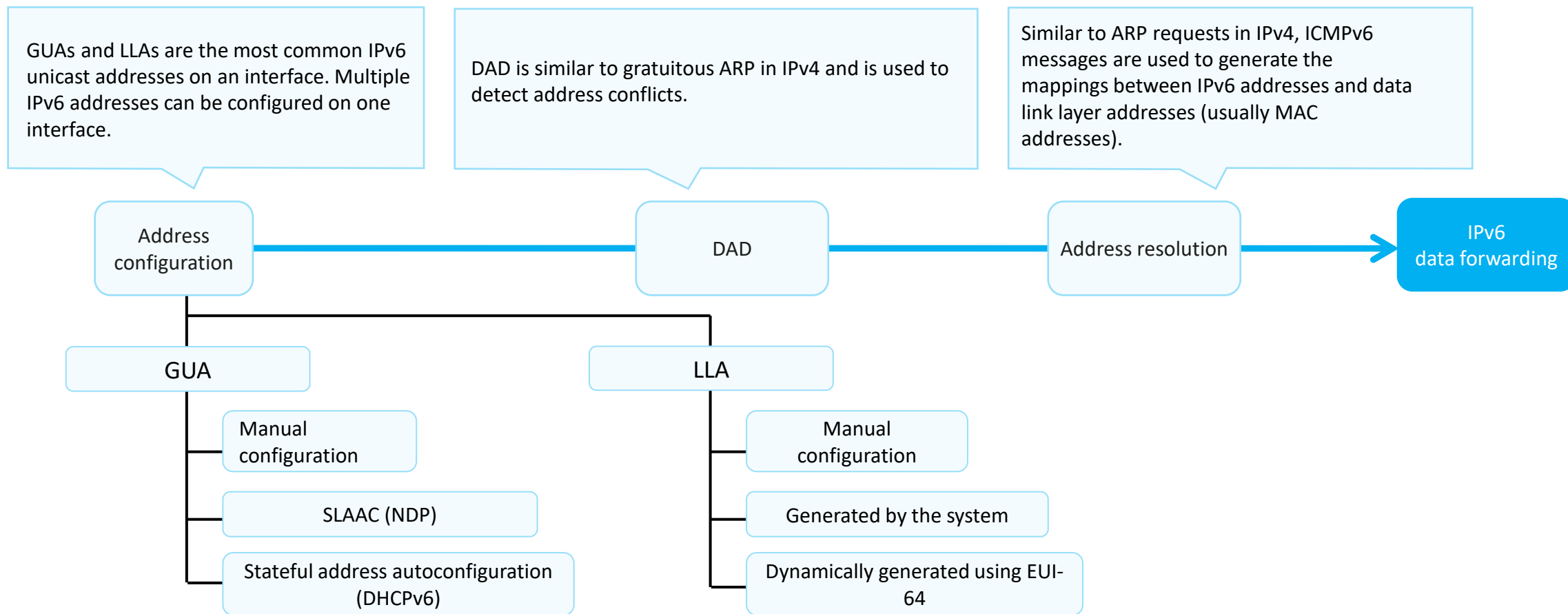
- The unicast IPv6 addresses and multicast addresses of hosts and routers are typically as follows:





Service Process of IPv6 Unicast Addresses

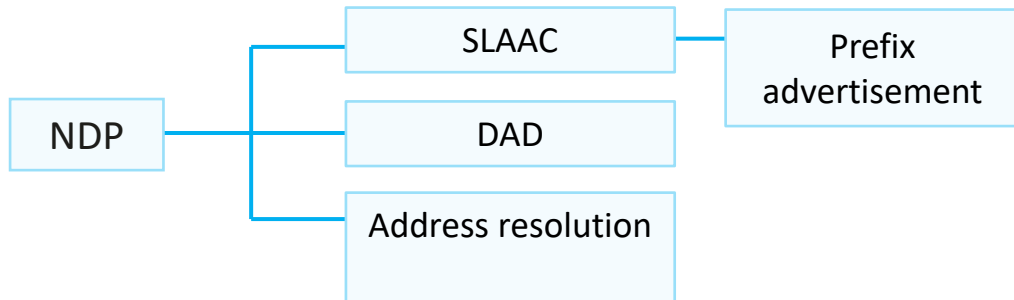
- Before sending IPv6 packets, an interface undergoes address configuration, DAD, and address resolution. During this process, the Neighbor Discovery Protocol (NDP) plays an important role.





NDP

- NDP is defined in RFC 2461, which was replaced by RFC 4861.
- NDP uses ICMPv6 messages to implement its functions.



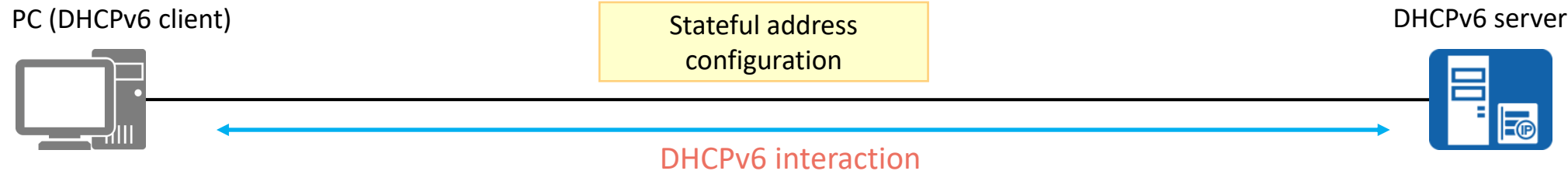
ICMPv6 messages used by NDP

ICMPv6 Type	Message Name
133	Router Solicitation (RS)
134	Router Advertisement (RA)
135	Neighbor Solicitation (NS)
136	Neighbor Advertisement (NA)

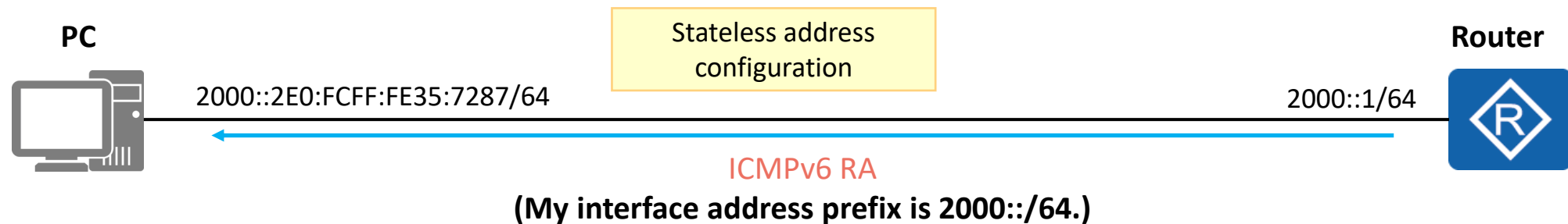
Mechanism	RS 133	RA 134	NS 135	NA 136
Address resolution			✓	✓
Prefix advertisement	✓	✓		
DAD			✓	✓



Dynamic IPv6 Address Configuration



- Through DHCPv6 message exchange, the DHCPv6 server automatically configures IPv6 addresses/prefixes **and other** network configuration parameters (such as DNS, NIS, and SNTP server addresses).

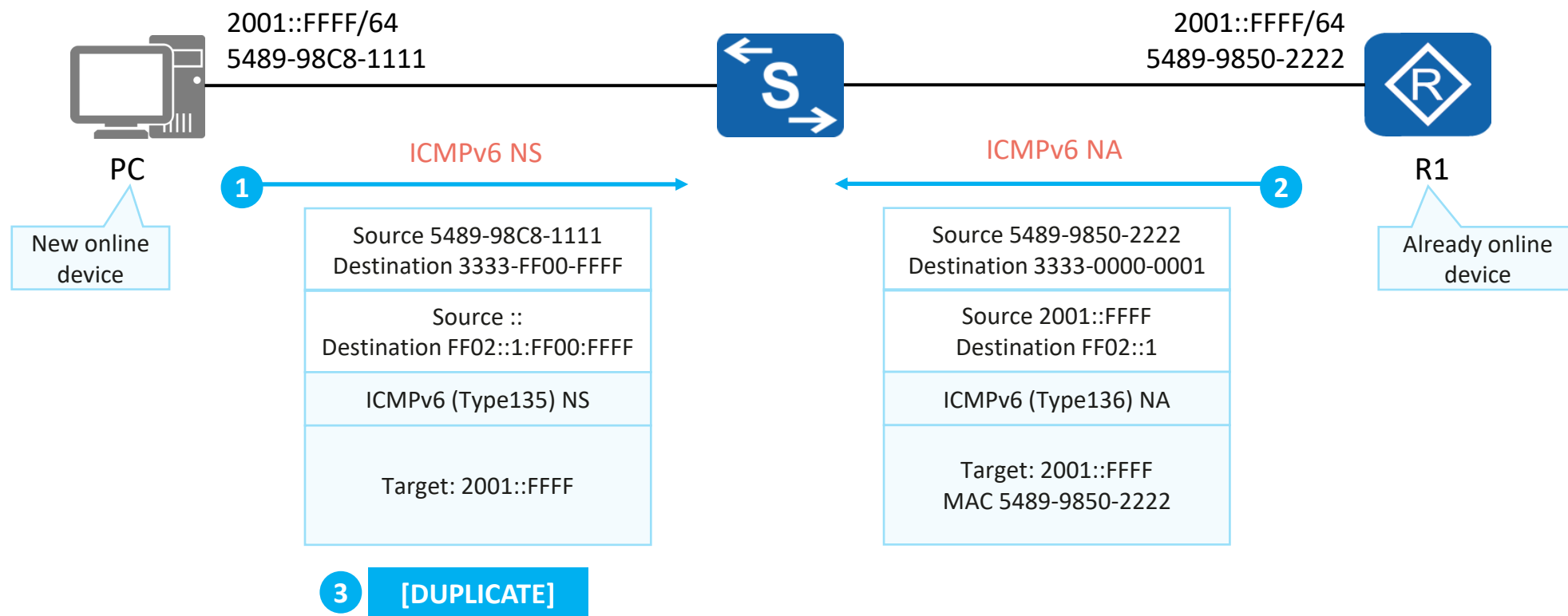


- The PC generates a unicast address based on the address prefix in the RA and the locally generated 64-bit interface ID (for example, using EUI-64).
- Only** IPv6 addresses can be obtained. Parameters such as NIS and SNTP server parameters cannot be obtained. DHCPv6 or manual configuration is required to obtain other configuration information.



DAD

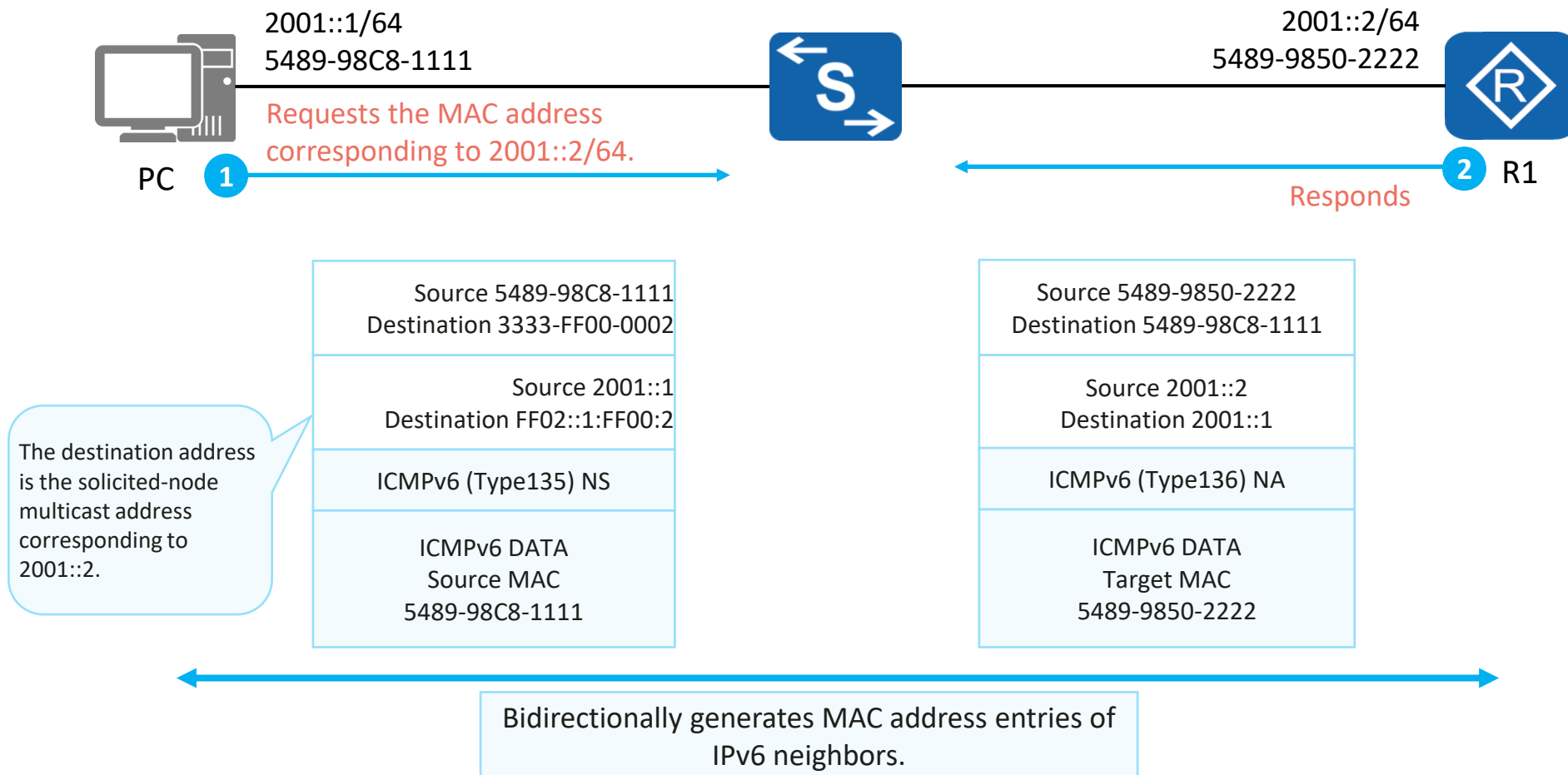
- Regardless of how an IPv6 unicast address is configured, a host or router:
 - Performs DAD through ICMPv6 messages.
 - Uses a unicast address only after passing the DAD procedure.





Address Resolution

- IPv6 uses ICMPv6 NS and NA messages to replace the address resolution function of ARP in IPv4.





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1. IPv6 Overview
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Basic IPv6 Configurations (1)

1. Enable IPv6.

```
[Huawei] ipv6
```

Enable the device to send and receive IPv6 unicast packets, including local IPv6 packets.

```
[Huawei-GigabitEthernet0/0/0] ipv6 enable
```

Enable IPv6 on the interface in the interface view.

2. Configure an LLA for the interface.

```
[Huawei-GigabitEthernet0/0/0] ipv6 address ipv6-address link-local
```

```
[Huawei-GigabitEthernet0/0/0] ipv6 address auto link-local
```

Configure an LLA for the interface manually or automatically in the interface view.

3. Configure a GUA for the interface.

```
[Huawei-GigabitEthernet0/0/0] ipv6 address { ipv6-address prefix-length | ipv6-address/prefix-length }
```

```
[Huawei-GigabitEthernet0/0/0] ipv6 address auto { global | dhcp }
```

Configure a GUA for the interface manually or automatically (stateful or stateless) in the interface view.



Basic IPv6 Configurations (2)

4. Configure an IPv6 static route.

```
[Huawei] ipv6 route-static dest-ipv6-address prefix-length { interface-type interface-number [ nexthop-ipv6-address ] | nexthop-ipv6-address } [ preference preference ]
```

5. Display IPv6 information on an interface.

```
[Huawei] display ipv6 interface [ interface-type interface-number | brief ]
```

6. Display neighbor entry information.

```
[Huawei] display ipv6 neighbors
```

7. Enable an interface to send RA messages.

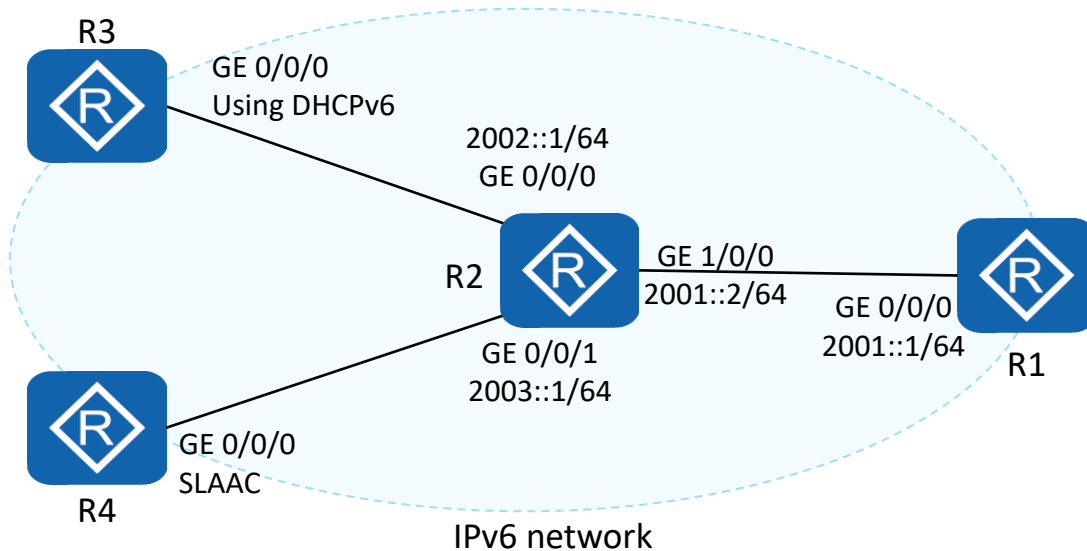
```
[Huawei-GigabitEthernet0/0/0] undo ipv6 nd ra halt
```

By default, a Huawei router's interfaces do not send ICMPv6 RA messages. In this situation, other devices on the links connected to the interfaces cannot perform SLAAC.

To perform SLAAC, you need to manually enable the function of sending RA messages.



Example: Configuring a Small IPv6 Network (1)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

1. Enable IPv6 globally and on related interfaces of R1, R2, R3, and R4, and enable the interfaces to automatically generate LLAs. The following uses R1 configurations as an example.

```
[R1]ipv6
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ipv6 enable
[R1-GigabitEthernet0/0/0]ipv6 address auto link-local
```

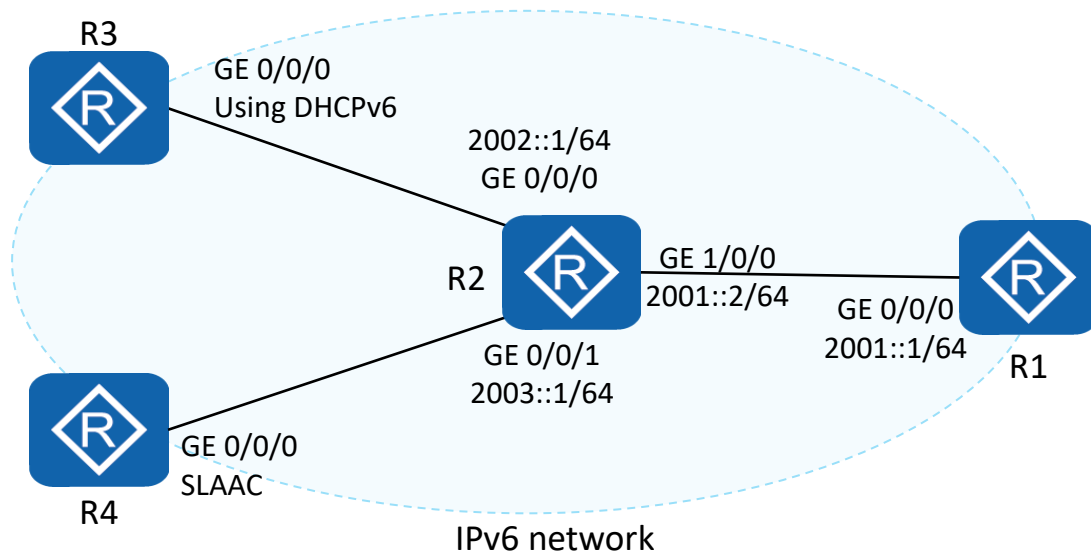
2. Configure static IPv6 GUAs on the related interfaces of R1 and R2.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ipv6 address 2001::1 64
```

```
[R2]interface GigabitEthernet 1/0/0
[R2-GigabitEthernet1/0/0]ipv6 address 2001::2 64
[R2-GigabitEthernet1/0/0]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ipv6 address 2002::1 64
[R2-GigabitEthernet0/0/0]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ipv6 address 2003::1 64
```




Example: Configuring a Small IPv6 Network (2)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

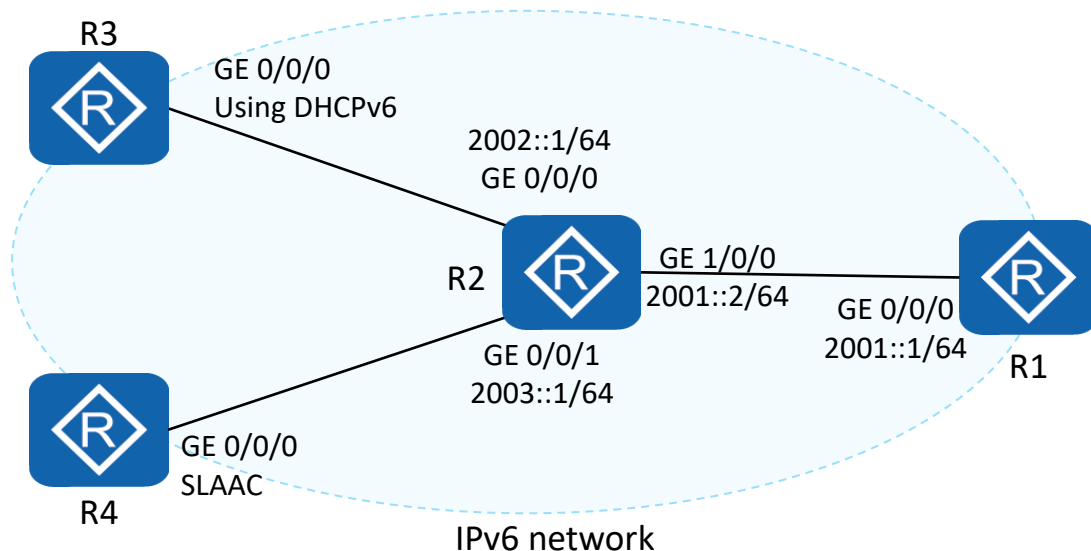
3. Configure R2 as a DHCPv6 server. Configure the related interface of R3 to obtain a GUA using DHCPv6.

```
[R2]dhcp enable
[R2]dhcpv6 pool pool1
[R2-dhcpv6-pool-pool1]address prefix 2002::/64
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]dhcpv6 server pool1
```

```
[R3]dhcp enable
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ipv6 address auto dhcp
```



Example: Configuring a Small IPv6 Network (3)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

4. Enable R2 to advertise RA messages. Enable R4 to obtain an address through SLAAC based on the RA messages sent by R2.

```
[R2]interface GigabitEthernet 0/0/1
```

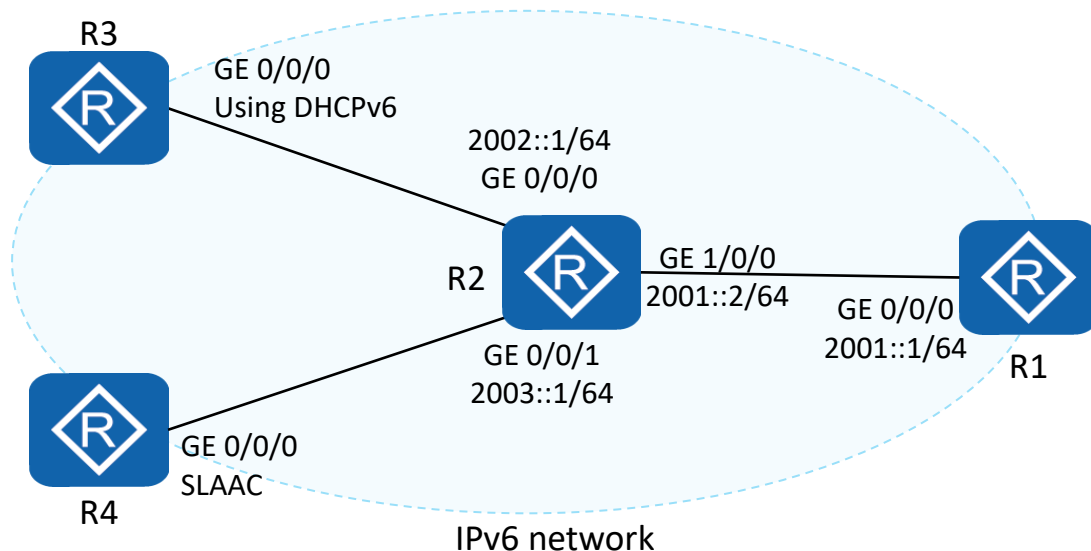
```
[R2-GigabitEthernet0/0/1]undo ipv6 nd ra halt
```

```
[R4]interface GigabitEthernet 0/0/0
```

```
[R4-GigabitEthernet0/0/0]ipv6 address auto global
```



Example: Configuring a Small IPv6 Network (4)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

5. Configure static routes on R4.

```
[R4]ipv6 route-static 2001:: 64 2003::1  
[R4]ipv6 route-static 2002:: 64 2003::1
```

6. Configure an aggregated static route on R1.

```
[R1]ipv6 route-static 2002:: 15 2001::2
```

7. Configure a default route on R3.

```
[R3]ipv6 route-static :: 0 2002::1
```



Quiz

1. What is the most abbreviated form of the IPv6 address 2001:0DB8:0000:0000:032A:0000:0000:2D70?
2. What is the process of SLAAC for IPv6 hosts?



Summary

Comparison	IPv6	IPv4
Address length	128 bits	32 bits
Packet format	A fixed 40-byte basic packet header+variable-length extension headers	A basic header containing the Options field to support extended features
Address type	Unicast, multicast, and anycast	Unicast, multicast, and broadcast
Address configuration	Static, DHCP, and SLAAC	Static and DHCP
DAD	ICMPv6	Gratuitous ARP
Address resolution	ICMPv6	ARP

The background of the slide features a blue-tinted image of several business professionals in a modern office. They are standing on a highly reflective floor, and their silhouettes are clearly visible. In the background, a city skyline with various skyscrapers is visible through large windows. The overall atmosphere is professional and corporate.

Thank You
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